EXP NO: 3(a)

AIM: Conduct an experiment to show data visualization using line plot

Description: Take any sample data either through csv file or data fetched directly through code.

PROGRAM:

```
import matplotlib.pyplot as cricket

Overs=list(range(5,51,5))

Indian_Score=[30,55,90,129,165,200,239,270,310,350]

Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]

cricket.plot(Overs,Indian_Score)

cricket.plot(Overs,Srilankan_Score)

cricket.show()

cricket.title("INDIA Vs SRILANKA")

cricket.xlabel("Overs")

cricket.ylabel("Score")

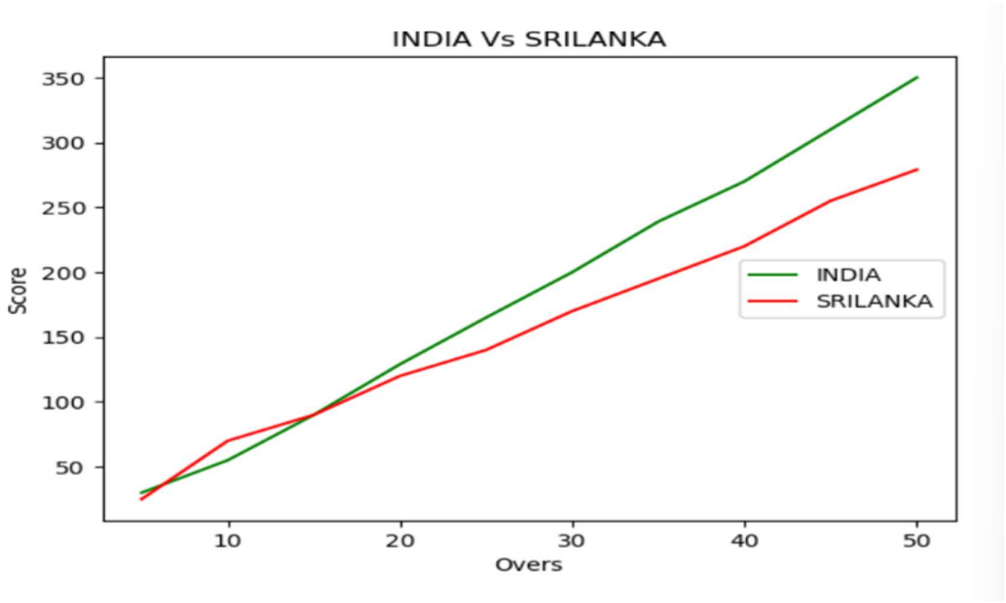
cricket.legend()

cricket.plot(Overs,Indian_Score,color="green",label="INDIA")

cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")

cricket.legend(loc="center right")
```

OUTPUT:



RESULT: Thus, an experiment to show data visualization using line plot is conducted.

EXP NO: 3(b)

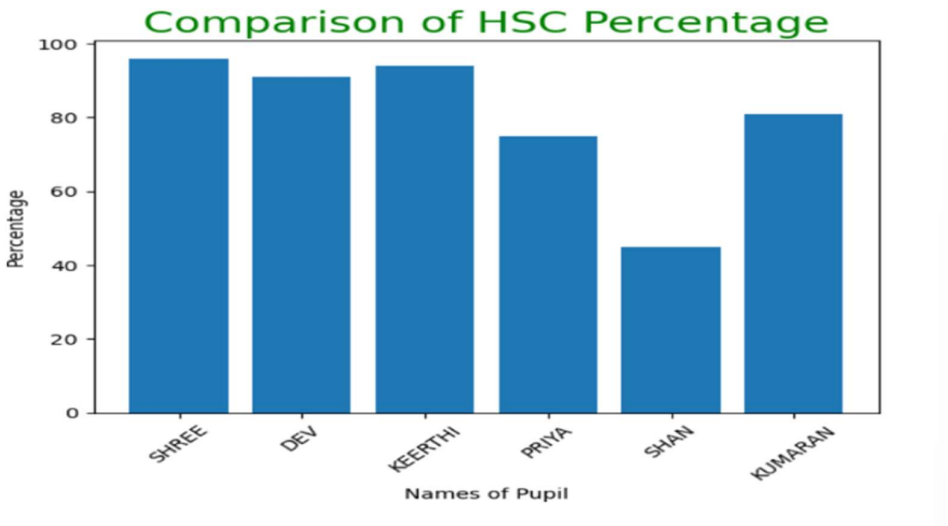
AIM: Conduct an experiment to show data visualization using bar chart.

DESCRIPTION: Take any sample data either through csv file or data fetched directly through code.

PROGRAM:

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

OUTPUT:



RESULT: Thus, the experiment to show the data visualization using bar chart is conducted.

EXP NO: 3(C)

AIM: Conduct an experiment to show data visualization using pie chart.

DESCRIPTION: Take any sample data either through csv file or data fetched directly through code.

PROGRAM:

```
import matplotlib.pyplot as election

# Election data

labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']

Votes = [315, 130, 245, 210]

colors = ['green', 'yellow', 'red', 'orange']

explode = (0.2, 0, 0, 0)

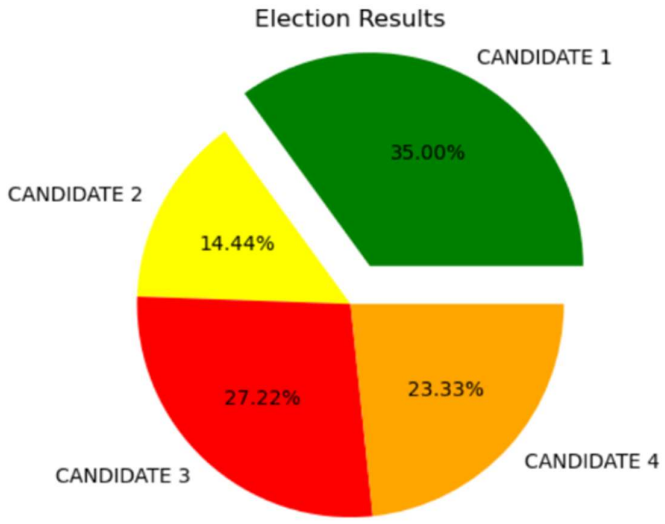
# Plotting the pie chart

election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')

election.title('Election Results')

election.show()
```

OUTPUT:



RESULT: Thus, an experiment to show data visualization using pie chart is conducted.